

## 1. Tool Recommendation: Power BI vs. Streamlit

After reviewing the weekly sales data for our departmental stores and understanding how often we'll be updating the dashboard, **Power BI** stands out as the better choice for this project.

**Power BI** is built specifically for business reporting — it's user-friendly, visually polished, and ideal for comparing sales across stores, departments, and time periods. It allows quick data refreshes whenever new weekly files arrive, supports automatic updates through OneDrive or Power BI Service, and provides a clean, mobile-friendly dashboard view that would appeal to our CEO.

On the other hand, **Streamlit** is a Python-based tool designed for developers. While it's great for creating custom web apps, it requires coding knowledge and manual maintenance every time new data is added. It's less practical for quick business reporting or for non-technical users who just need to view updated sales metrics.

### Recommendation:

For our scale of operations — multiple stores, weekly sales uploads, and executive reporting — **Power BI is the most efficient, scalable, and visually appealing option.** It ensures consistency, automation, and accessibility without needing ongoing technical intervention.

## 2. Data Upload Recommendations

After adding the new week's data for October 26, 2012, into the Power BI dashboard, I noticed that the refresh took about two to three minutes. While that's fine for now, it's easy to see that as more weeks get added, the process will start slowing down because Power BI reloads every single file each time we refresh. It also relies on manual updates and local performance, which isn't sustainable in the long run.

To make things smoother, I'd suggest keeping all the weekly sales files in one shared folder on OneDrive or SharePoint and connecting Power BI directly to it. Once the report is published to the Power BI Service, we can set up automatic refreshes so that every time new data is added, the dashboard updates on its own. This way, no one has to manually refresh it, and the numbers will always be up to date for the management team.

In the future, we can take it a step further by setting up incremental refresh or moving the data into a proper SQL database. That would make the process even faster and help the dashboard handle larger volumes of data as the company continues to grow.